THARUN DERANGULA

Full-Stack Software Engineer | AI/ML Integration | Distributed Systems

San Francisco Bay Area, CA | +1 (408) 652-9472 | dtharun209@gmail.com | LinkedIn | GitHub | Portfolio

SUMMARY

Full-stack engineer with 4 years of experience building and scaling production systems across frontend, backend, and cloud. Strong in distributed systems, event-driven services, and high-traffic APIs serving millions of users, with recent focus on AI features using RAG, vector retrieval, and LLM integration at Walmart.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript (ES6+), Python, Go, SQL
Frontend: React, Next.js (SSR), Redux, Tailwind CSS, Jest, React Testing Library, Cypress
Backend: Node.js, Express.js, Spring Boot, REST, GraphQL, gRPC, Microservices, Kafka, Redis
AI/ML: LLM Integration, RAG, pgvector, Vector Retrieval, LangChain, OpenAI API, Anthropic API
Cloud & DevOps: GCP, AWS (EC2, S3, DynamoDB, CloudWatch), Docker, Kubernetes, Terraform, Helm, GitHub Actions, Prometheus, Grafana
Databases: PostgreSQL, MongoDB, Redis, Snowflake, Elasticsearch, DynamoDB
AI Dev Tools: Claude Code, GitHub Copilot, Cursor, Codex
Practices: System Design, Event-Driven Architecture, Domain-Driven Design, TDD, CI/CD, Agile/Scrum

EXPERIENCE

Software Engineer, Walmart Global Tech — Sunnyvale, CA April 2025 – Present

- Shipped customer-facing features for Sparky, Walmart's AI shopping agent, delivering conversational shopping and cart workflows across web and mobile with **React**, **Next.js**, and **TypeScript**, partnering with PM, design, and ML teams.
- Improved AI recommendation relevance by 20% in A/B testing, lifting add-to-cart conversion on recommendation surfaces, by integrating Walmart's Wallaby **LLM** via the Element ML Platform using **RAG** with **pgvector**-based vector retrieval.
- Architected cart auto-build and intent-routing microservices in **Java**, **Spring Boot**, **Kafka**, and **PostgreSQL** processing millions of daily shopping events, with idempotent retries, dead-letter queues, and near real-time **GCP Pub/Sub** pipelines for inventory and fulfillment.
- Cut p95 latency by 35% on high-traffic recommendation endpoints through **Redis** caching, connection pooling, and **PostgreSQL** index and query-plan tuning, holding sub-50ms median latency through seasonal and Rollback traffic spikes.
- Enabled zero-downtime releases and reduced deploy effort by half by automating infrastructure with **Docker**, **Kubernetes**, **Terraform**, **WCNP**, and **GitHub Actions**, backed by **Datadog** and **Grafana** observability.
- Secured shopping and cart APIs with **JWT** and **OAuth 2.0** authentication and role-based access control, while maintaining 85%+ test coverage with **Jest** and **React Testing Library**.
- Authored technical design docs and led design reviews for cart auto-build and intent-routing services, driving alignment across PM, ML, and platform teams.
- Shortened delivery cycles by adopting **Claude Code** and **GitHub Copilot** for scaffolding, refactors, and test generation across frontend and backend codebases.

Software Engineer, Kennesaw State University — Kennesaw, GA January 2024 – December 2024

- Built KSU's Campus Hub with **React**, **Next.js**, and **TypeScript**, consolidating Owl Express, D2L Brightspace, DegreeWorks, and KSUMail into one responsive platform serving 40,000+ students and faculty.
- Improved dashboard load time by 30% by developing **Node.js** and **Spring Boot REST** APIs with **Redis** caching and query tuning across Ellucian Banner and Degree Works integrations.
- Reduced manual data correction by 40% by building **Java** and **Spring Boot** services over **PostgreSQL** with **REST**-webhook reconciliation for enrollment, grades, and financial aid data.
- Shipped low-disruption releases during peak registration by deploying on **AWS EC2** and **S3** with **Docker** and **GitHub Actions CI/CD**, cutting deployment cycle time by half.
- Caught regressions earlier by maintaining 80%+ coverage with **Jest** and **React Testing Library** and collaborating with UITS teams in Agile sprints through design and code reviews.

Software Engineer, HCL Tech — India May 2020 – December 2022

- Built invoice tracking and payment workflows for enterprise banking customers across regions with **React** and **TypeScript** frontends backed by **Java** and **Spring Boot** APIs.
- Improved peak-period API responsiveness by optimizing **Java** and **Spring Boot** services on **PostgreSQL** and **Redis** with caching and connection pooling.
- Engineered event-driven payment processing with **Kafka**, **Java**, and **Spring Boot** using idempotent retries and dead-letter queues for reliable transaction handling under high load.
- Cut batch processing time by 45% for analytics and compliance reporting by building ETL pipelines with **Python** and **GCP BigQuery** for downstream finance teams.
- Took releases from days to hours and cut production defects by 40% by modernizing legacy backends into microservices on **Docker**, **Terraform**, **GCP Cloud Run**, and **GKE** with **GitHub Actions CI/CD** and **TDD** (**JUnit**, **pytest**).
- Cut mean time to resolution by 30% by adding structured logging, alerting, and dashboards across payment services for faster root-cause analysis.
- Mentored two junior engineers and drove code reviews and API design discussions across the payments platform.

EDUCATION

Master of Science, Information Technology | Kennesaw State University 2023 – 2024
Bachelor of Science, Computer Science & Engineering | Lovely Professional University 2018 – 2022
Certifications: Google Professional Cloud Architect | AWS Certified Solutions Architect | Microsoft Generative AI Engineering Professional